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Distinct genomic routes underlie transitions to specialised symbiotic lifestyles in deep-sea annelid worms

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Bacterial symbioses allow annelids to colonise extreme ecological niches, such as hydrothermal vents and whale falls. Yet, the genetic principles sustaining these symbioses remain unclear. Here, we show that different genomic adaptations underpin the symbioses of phylogenetically related annelids with distinct nutritional strategies. Genome compaction and extensive gene losses distinguish the heterotrophic symbiosis of the bone-eating worm *Osedax frankpressi* from the chemoautotrophic symbiosis of deep-sea Vestimentifera. *Osedax*'s endosymbionts complement many of the host's metabolic deficiencies, including the loss of pathways to recycle nitrogen and synthesise some amino acids. *Osedax*'s endosymbionts possess the glyoxylate cycle, which could allow more efficient catabolism of bone-derived nutrients and the production of carbohydrates from fatty acids. Unlike in most Vestimentifera, innate immunity genes are reduced in *O. frankpressi*, which, however, has an expansion of matrix metalloproteases to digest collagen. Our study supports that distinct nutritional interactions influence host genome evolution differently in highly specialised symbioses.

Symbioses have shaped life on Earth, from the origin of the eukaryotic cell to the formation of biodiversity hotspots such as coral reefs[1,2]. Animal chemosynthetic symbioses, where bacteria convert inorganic compounds to organic matter, are ubiquitous in marine habitats[3] and fuel some of the most productive communities, such as those around hydrothermal vents[4]. Siboglinid worms (Annelida) often dominate deep-sea chemosynthetic environments through symbioses with environmentally acquired bacteria[5,6] that adults harbour within a specialised organ called a trophosome[7]. Despite their ecological importance, the host's genetic traits sustaining these symbioses have only been studied in Vestimentifera[8–10], one of the four main lineages in Siboglinidae (Fig. 1a). The genomes of *Lamellibrachia luymesi*[8], *Paraescarpia echinospica*[9], *Riftia pachyptila*[10] and *Ridgeia piscesae*[11] have revealed a complex molecular interplay between Vestimentifera and their endosymbionts to fulfil their nutritional demands[12]. For example, the hosts have lost genes involved in amino acid biosynthesis[8,10,11] and carbohydrate catabolism[9] but expanded gene families involved in nutrient transport[8], gas exchange[8–10,13,14], innate immunity[9,11] and

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